RETAIL & E-COMMERCE SALES PERFORMANCE

TECHNICAL ANALYSIS REPORT

Analytical Period: 1 January 2026 to 31 December 2026

Prepared from the supplied Excel workbook and completed Sales Performance Dashboard

Technical framework: Source Data → Preparation → Methodology → Validation → Analysis → Findings → Recommendations → Conclusion

Document Control and Scope

- Project: Retail & E-commerce Sales Performance Dashboard.
- Primary source workbook: Retail & E-commerce Sales Performance Dashboard.xlsx.
- Primary transaction sheet: Sales_Data.
- Supporting analytical sheet: Supporting_Pivots.
- Dashboard design/reference sheet: Dashboard_Plan.
- Presentation sheet: Sales_Dashboard.
- Source population: 2,000 transaction/order records.
- Reporting period supported by the source data: 01 January 2026 to 31 December 2026.
- Reporting constraint: This document deliberately contains no tables. All technical results are communicated through narrative analysis, structured headings, and bullet-point findings.

Report Structure

1. Executive Summary

This technical report documents the analysis of the supplied retail and e-commerce sales dataset and the resulting Sales Performance Dashboard. The transaction-level source contains 2,000 records covering the full 2026 calendar year. Total recorded revenue is $2,452,084.30, total quantity sold is 5,552 units, and average revenue per order is $1,226.04.

Revenue performance is led by the North region at $561,753.72, followed closely by Central at $556,705.18. At product-category level, Computers is the dominant category with $1,346,491.26, representing 54.9% of total revenue. Online is the leading sales channel at $791,536.99, while Returning customers generate $1,683,325.30.

Monthly revenue reaches its highest level in November at $274,368.31, while July records the lowest monthly revenue at $150,713.14. The final quarter contributes $650,608.89, equivalent to 26.5% of annual revenue. Among individual products, Gaming Laptop is the highest-revenue product at $432,576.06.

The dashboard is therefore technically effective as a descriptive management-reporting tool because it combines KPI cards, time-series analysis, regional comparison, product-category performance, sales-channel contribution and top-product ranking with interactive filters. The analysis also identifies areas for management attention, particularly product concentration, regional performance differences, channel mix, and the need to monitor discounting and customer retention.

2. Introduction and Business Context

The project converts transaction-level retail and e-commerce sales records into a structured analytical dashboard for performance monitoring and decision support. The supplied workbook separates the raw transactional dataset from supporting pivot calculations and the final dashboard presentation layer.

The dashboard is designed to answer practical business questions: how much revenue has been generated, how many orders were processed, how many units were sold, what the average order value is, how revenue changes over time, which regions and product categories perform best, which sales channels contribute most, and which individual products generate the highest revenue.

3. Analytical Objectives

The technical analysis was structured around the following objectives:

- Validate the transaction-level dataset before analytical aggregation.
- Calculate total revenue, total orders, total quantity sold and average order value.
- Analyse revenue by month to identify seasonality, peaks and troughs.
- Measure revenue contribution by geographic region.
- Evaluate product-category revenue performance.
- Measure sales-channel contribution to total revenue.
- Identify the top five products by revenue.
- Assess customer-type contribution between new and returning customers.
- Provide an interactive management reporting structure through dashboard filters.
- Translate descriptive patterns into evidence-based business recommendations.

4. Data Source and Dataset Structure

The primary source is the Sales_Data worksheet. It contains 12 fields covering order identification, transaction date, geography, sales channel, customer type, product classification, individual product, salesperson, quantity, unit price, discount and revenue.

The transaction population contains 2,000 records. Order IDs are populated across the dataset and the source contains 2,000 unique orders. The Date field spans 1 January 2026 through 31 December 2026, with 363 distinct dates represented.

The Dashboard_Plan worksheet defines the intended analytical requirements and maps each dashboard element to its source field. The Supporting_Pivots worksheet contains the KPI calculations and chart-supporting aggregations used by the dashboard.

The Sales_Dashboard worksheet provides the presentation layer. The supplied dashboard image shows filters for Region, Years, Quarters, Months, Days, Product Category, Sales Channel and Customer Type, enabling users to dynamically restrict the displayed analytical results.

5. Analytical Methodology

The project follows a descriptive analytics workflow:

14. Source inspection and field interpretation.
15. Data completeness and consistency checks.
16. Metric definition and KPI calculation.
17. Aggregation of revenue across time, geography, category, channel and product.
18. Ranking of dimensions by revenue contribution.
19. Trend analysis using monthly revenue.
20. Dashboard construction from supporting pivot-style calculations.
21. Cross-checking displayed KPI values against the transaction-level dataset.
22. Interpretation of findings and formulation of recommendations.

5.1 KPI Definitions

- Total Revenue is the sum of the Revenue field: $2,452,084.30.
- Total Orders is the count of unique Order IDs: 2,000.
- Quantity Sold is the sum of Quantity: 5,552 units.
- Average Order Value is calculated as total revenue divided by total orders: $1,226.04.
- Revenue by Month is the aggregation of Revenue by calendar month derived from Date.
- Revenue by Region is the aggregation of Revenue by Region.
- Revenue by Product Category is the aggregation of Revenue by Product Category.
- Revenue by Sales Channel is the aggregation of Revenue by Sales Channel.
- Top 5 Products by Revenue is the descending ranking of Product based on total Revenue.

6. Data Preparation and Quality Assessment

6.1 Completeness

The supplied Sales_Data worksheet is complete across all 12 fields for the 2,000 records analysed. No missing values were detected in Order ID, Date, Region, Sales Channel, Customer Type, Product Category, Product, Salesperson, Quantity, Unit Price, Discount or Revenue.

6.2 Duplicate Order Validation

Order ID contains 0 duplicated values when checked at the transaction-record level. The source therefore supports the dashboard's use of Order ID as an order-counting field without evidence of duplicate order identifiers in the supplied dataset.

6.3 Numeric Validation

Quantity contains no non-positive values, Unit Price contains no negative values, and Discount contains no values outside the 0-to-1 range. A recalculation of revenue using Unit Price × Quantity × (1 − Discount) produced a maximum absolute difference of $0.0000000000 from the supplied Revenue field, which is effectively a rounding-level reconciliation.

6.4 Date Validation

The Date field is recognised as a valid date field and covers 01 January 2026 through 31 December 2026. This supports time grouping into years, quarters, months and days as implemented in the dashboard.

7. Dashboard Architecture and Technical Implementation

The dashboard uses a layered design. The top layer presents four management KPIs: Total Revenue, Total Orders, Quantity Sold and Average Order Value. The middle and lower analytical layers provide time, geographic, category, channel and product perspectives.

Interactive slicer/filter controls are positioned around the dashboard. Region and Product Category filters support geographic and portfolio analysis, while Sales Channel and Customer Type filters support commercial segmentation. Date controls for Years, Quarters, Months and Days provide temporal drill-down.

The supporting pivot architecture separates KPI calculations from chart-supporting aggregations. This approach improves maintainability because the dashboard visuals can be refreshed from a consistent analytical layer rather than requiring separate calculations inside each visual.

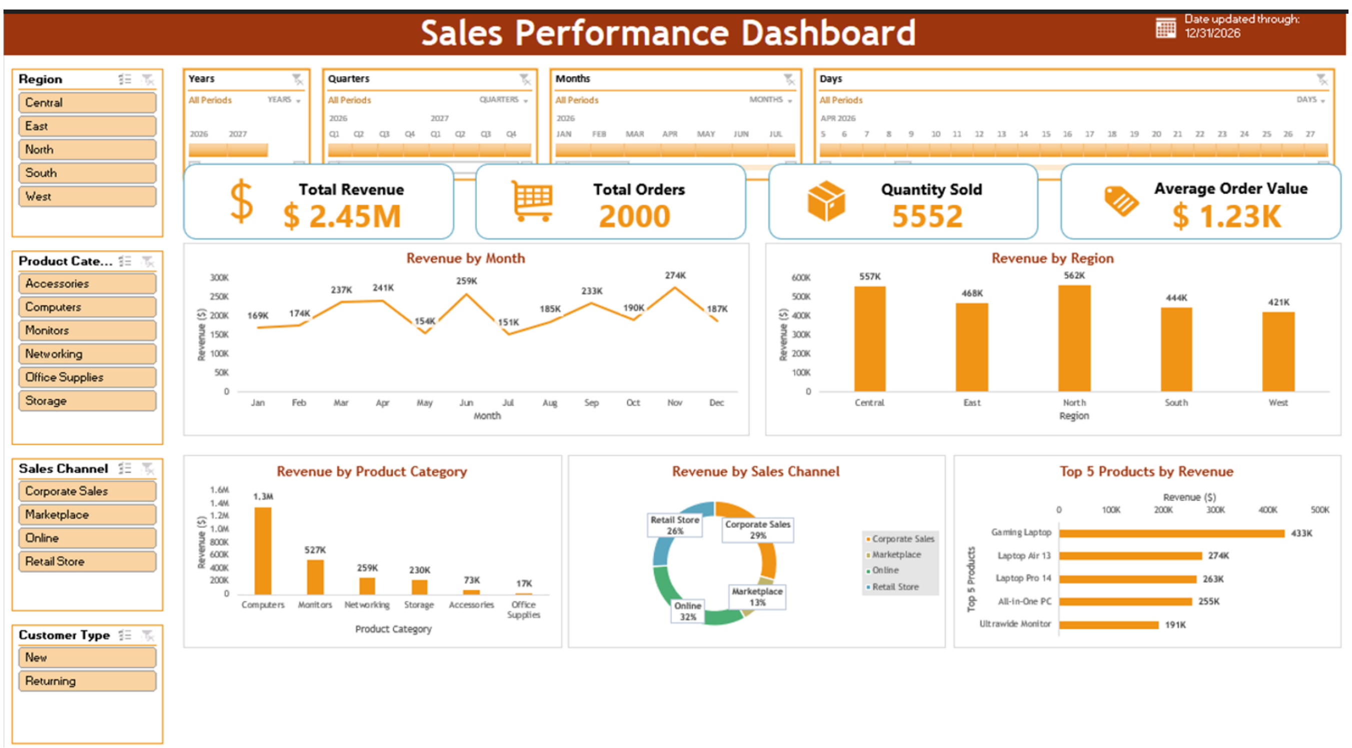

Figure 1. Supplied Retail & E-commerce Sales Performance Dashboard.

8. Detailed Analytical Findings

8.1 Overall Revenue and Order Performance

The business generated $2,452,084.30 from 2,000 orders and sold 5,552 units during 2026. The resulting average order value is $1,226.04. These values reconcile with the KPI calculations in the Supporting_Pivots worksheet and the dashboard.

8.2 Monthly Revenue Trend

Revenue shows substantial month-to-month movement. November is the strongest month at $274,368.31, followed by March at $236,738.70 and April at $241,034.34. The lowest month is July at $150,713.14, followed by May at $153,642.43.

The difference between the highest and lowest months is $123,655.17. The final quarter contributes 26.5% of annual revenue, with November representing the annual peak. This pattern indicates that sales planning should account for meaningful seasonal variation rather than assume a uniform monthly revenue profile.

8.3 Regional Performance

- North: $561,753.72, representing 22.9% of total revenue.
- Central: $556,705.18, representing 22.7% of total revenue.
- East: $467,808.85, representing 19.1% of total revenue.
- South: $444,494.30, representing 18.1% of total revenue.
- West: $421,322.25, representing 17.2% of total revenue.

North is the highest-performing region at $561,753.72, marginally ahead of Central at $556,705.18. East, South and West contribute progressively lower revenue. The relatively close performance of North and Central suggests that the business has two major geographic revenue centres, while the gap between the highest and lowest regions provides an opportunity for targeted regional improvement.

8.4 Product Category Performance

- Computers: $1,346,491.26, representing 54.9% of total revenue.
- Monitors: $526,744.59, representing 21.5% of total revenue.
- Networking: $259,475.31, representing 10.6% of total revenue.
- Storage: $229,837.63, representing 9.4% of total revenue.
- Accessories: $72,895.26, representing 3.0% of total revenue.
- Office Supplies: $16,640.26, representing 0.7% of total revenue.

Computers is the dominant category at $1,346,491.26, accounting for 54.9% of total revenue. Monitors is second at $526,744.59, while Networking and Storage contribute $259,475.31 and $229,837.63 respectively.

The three largest categories collectively generate $2,132,711.16, equivalent to 87.0% of annual revenue. This concentration means category strategy, inventory planning and promotional decisions should pay particular attention to the leading technology categories.

8.5 Sales Channel Performance

- Online: $791,536.99, representing 32.3% of total revenue.
- Corporate Sales: $718,745.01, representing 29.3% of total revenue.
- Retail Store: $631,399.37, representing 25.7% of total revenue.
- Marketplace: $310,402.93, representing 12.7% of total revenue.

Online is the leading channel, followed by Corporate Sales and Retail Store. Marketplace is the smallest of the four channels. The channel distribution indicates that digital and direct commercial channels are central to revenue generation, while Marketplace remains a smaller but measurable contribution.

8.6 Top Product Performance

- Gaming Laptop: $432,576.06.
- Laptop Air 13: $273,532.50.
- Laptop Pro 14: $263,486.40.
- All-in-One PC: $254,994.00.
- Ultrawide Monitor: $190,959.45.

The top five products generate $1,415,548.41, representing 57.7% of total revenue. Gaming Laptop is the clear leading product, generating materially more revenue than the second-ranked Laptop Air 13. This indicates meaningful product-level concentration and suggests that availability and demand planning for high-value products should be closely monitored.

8.7 Customer-Type Performance

- Returning customers: $1,683,325.30, representing 68.6% of total revenue.
- New customers: $768,759.00, representing 31.4% of total revenue.

Returning customers contribute substantially more revenue than new customers. This indicates that retention and repeat purchasing are important drivers of the observed sales base and should be incorporated into commercial planning.

9. KPI and Visual Interpretation

The Total Revenue KPI communicates the overall financial scale of the analysed sales population. The Total Orders KPI measures transaction volume, while Quantity Sold measures physical unit movement. Average Order Value provides a complementary measure of revenue generated per order and helps distinguish changes in revenue caused by order volume from changes caused by order size.

The Revenue by Month visual is appropriate for identifying seasonality and short-term changes. The Revenue by Region visual provides direct comparison across the five geographic regions. The Product Category chart exposes portfolio concentration, while the Sales Channel chart shows the contribution of each route to market. The Top 5 Products visual is particularly useful for identifying high-value products requiring commercial and inventory attention.

The dashboard's filters add an interactive analytical layer. For example, users can isolate a region, category or sales channel and then observe how the selected segment changes the KPI and chart outputs. This supports drill-down analysis, but filtered values should always be interpreted as subset results rather than company-wide totals.

10. Key Technical Observations

- Total revenue is $2,452,084.30 across 2,000 orders.
- Average order value is $1,226.04, while 5,552 units were sold.
- North is the highest-revenue region at $561,753.72.
- Computers is the largest product category and contributes 54.9% of revenue.
- Online is the leading sales channel with $791,536.99.
- Gaming Laptop is the highest-revenue individual product at $432,576.06.
- November is the strongest month at $274,368.31; July is the weakest at $150,713.14.
- Returning customers generate 68.6% of revenue, materially exceeding the contribution from new customers.

- The source data is complete for all required fields and passes the principal numeric and date validation checks.
- The dashboard architecture aligns with the analytical requirements documented in the Dashboard_Plan worksheet.

11. Recommendations

11.1 Protect and Scale High-Value Product Lines

Prioritise Gaming Laptop and the other leading products in demand forecasting, inventory availability and promotional planning. Because the top products account for a substantial share of revenue, stock-outs or pricing issues in these products could have disproportionate revenue effects.

11.2 Strengthen the Lower-Performing Regions

North and Central are the strongest regions, while West is the lowest. Management should assess customer coverage, sales activity, product availability and channel effectiveness in West and South to identify practical growth constraints.

11.3 Optimise Channel Strategy

Online is the leading channel and should continue to receive strategic attention. At the same time, Marketplace has the smallest contribution and should be assessed for conversion, product assortment, pricing competitiveness and customer acquisition efficiency before additional investment is made.

11.4 Use Seasonal Planning

The strong November and June performance and the weaker May and July results indicate seasonality. Sales forecasts, inventory procurement and campaign calendars should therefore incorporate month-specific expectations rather than relying only on annual targets.

11.5 Strengthen Customer Retention

Returning customers generate the majority of revenue. The business should therefore maintain structured retention initiatives, repeat-purchase campaigns and customer-value monitoring to protect the existing revenue base while continuing to acquire new customers.

11.6 Monitor Discount Impact

Discount is a core source field and is already incorporated into the Revenue calculation. Future dashboard iterations should consider adding a discount-performance view that evaluates whether higher discount levels generate sufficient incremental revenue or volume to justify margin sacrifice.

11.7 Maintain Dashboard Governance

The dashboard should be refreshed from the transaction source and supporting pivot calculations using a controlled refresh process. KPI definitions should remain documented, and any changes to source fields, filter logic or aggregation rules should be version-controlled.

12. Limitations and Data Governance

The analysis is descriptive and is based exclusively on the supplied workbook and dashboard. It identifies observed patterns but does not establish causal relationships. For example, the report cannot determine from the dataset alone why November is the strongest month.

The Revenue field is treated as the final sales amount supplied by the workbook. The report does not estimate profit, gross margin, customer acquisition cost, fulfilment cost or channel profitability because those measures are not provided.

Salesperson performance is present in the source data but is not included as a dedicated visual in the supplied dashboard. It can therefore support additional analysis, but it should not be interpreted as a current dashboard KPI unless explicitly added to the dashboard.

The dashboard date indicator states that the data is updated through 12/31/2026, which aligns with the maximum Date value in the transaction data. This provides consistency between the displayed update date and the underlying reporting period.

13. Conclusion

The retail and e-commerce sales analysis demonstrates a revenue base of $2,452,084.30 across 2,000 orders and 5,552 units during 2026. Performance is concentrated around the North and Central regions, the Computers category, Online sales and a small group of high-revenue products.

Monthly performance also varies materially, with November reaching $274,368.31 and July falling to $150,713.14. Returning customers form the larger revenue base, reinforcing the importance of retention and repeat purchasing.

From a technical reporting perspective, the workbook provides a strong foundation for recurring sales-performance monitoring. The source data is complete, the core revenue calculation reconciles, the supporting pivot layer is aligned with the documented dashboard requirements, and the final dashboard provides an effective interactive presentation layer.

The recommended next step is to operationalise the dashboard as a recurring management reporting process, with controlled data refreshes, documented KPI definitions, ongoing monitoring of product and regional concentration, and additional profitability-oriented metrics when the required cost data becomes available.

Appendix A: Technical Source Traceability

- Sales_Data: transaction-level source used for all primary calculations.
- Dashboard_Plan: documented field meanings, validation checks and dashboard requirements.
- Supporting_Pivots: KPI calculations and chart-supporting aggregations.
- Sales_Dashboard: presentation layer containing the completed dashboard.
- Supplied screenshot: visual reference used to document the final dashboard layout and interactive controls.

Appendix B: Validation Summary

- Record count: 2,000.
- Unique Order IDs: 2,000.
- Missing values across required source fields: 0.
- Non-positive quantities: 0.
- Negative unit prices: 0.
- Invalid discount values outside 0–1: 0.
- Maximum absolute revenue reconciliation difference: $0.0000000000.
- Date range: 01 January 2026 to 31 December 2026.
- Distinct dates represented: 363.